

Multi-Agent Mentor-Student Simulation — Assignment Solution

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Assignment. Two LLM agents run a 10-lesson “Prompt Engineering” course: a **mentor** checks *application* of each skill (not recall), using questions only — no files/screenshots shown to the student. The **student** sometimes bluffs (claims practice it didn’t do). The mentor must catch this from conversation alone.

Architecture

- **Two mirrored contexts, not one shared chat** — mentor_ctx/student_ctx are separate message lists; the relay forwards each turn into the other’s context. Fresh per lesson; cross-lesson memory rides durable tools (mentor_ledger, student_practice_log), not raw chat history.
- **Fixed-phase flow + forced gate:** greet/segue → open (introduce + verify) → apply (transfer scenario) → probe (adaptive follow-ups) → forced advance_decision. Guarantees one structured verdict (PASS/BLUFF_SUSPECTED) per lesson.
- **Bluff detection, three levers:** (1) *canary facts* — every rule uses a made-up term (“1-task-1-verb rule”, “Contrast rule”, ...) absent from any model’s pretraining; (2) answer_key.json — private to the mentor, canary facts + expected application points; (3) student_practice_log.json — on bluff lessons (3, 6, 9) nothing is written, so practice_read returns null and the student must fabricate or admit it.

Conversational quality (feedback fixes)

Two changes so the transcript reads like a real chat rather than a lecture:

- **A user never replies with long messages.** Brevity is now the student prompt’s top rule (1–2 sentences), with a STUDENT_MAX_TOKENS backstop. Longest student reply: **70 words**, down from ~200-word paragraphs.
- **Never open a conversation with a lesson and a task** — that loses the customer. The session now opens with a greeting exchange, and every lesson eases in with a segue before the concept:

```
Mentor: Hey there! How's it going? Have you had a chance to dive into any practice lately?
Student: Hey! I'm doing well, thanks! Yes, I did some practice recently.
Mentor: Great to hear! Today, I want to talk about the "1-task-1-verb" rule...
```

Models & tools

Mentor/student both openai/gpt-4o-mini (OpenRouter); per-role models are supported (MENTOR_MODEL/STUDENT_MODEL), and the student runs with reasoning disabled (kills empty-message blank-outs). Judge is a deterministic precision/recall calculator (no LLM). Tools: ledger_read/ledger_write (mentor memory), practice_read (student’s only source of concrete detail), practice_simulator (deterministic, seeded practice-outcome generator), advance_decision (the hard gate), judge (post-run scoring).

Result (best run — run_018, 10/10 lessons, seed 7)

```
precision=1.0  recall=1.0      TP=3 FP=0 FN=0
caught=[3, 6, 9]  missed=[]    (bluffs forced on 3, 6, 9)
```

Across 5 seeds: **recall 1.0 every run** — all three forced bluffs caught every time; precision ranged **0.6–1.0** (0–2 honest lessons over-flagged). The previous version reported recall 0.667 on a different model (deepseek-v4-pro), so this is not a like-for-like comparison.

Reflection

Making the student terse traded directly against detection: shorter replies give the mentor less concrete detail to judge, so it began over-flagging *honest* students (precision fell to 0.6 until a “brevity is not evasion” rule in the mentor prompt pulled it back). There is a real tension between “text like a real user” and “give enough evidence that you practised” — the two fixes above are not free. The honest ceiling of conversation-only detection still stands: a strong model told to bluff can manufacture *plausible* friction, not just a smooth denial, and nothing here separates manufactured friction from real friction once both are equally specific. Mentor prompt is at 5 iterations, student at 4 (early mentors accepted a correct transfer answer alone, or condemned a thin-but-honest first answer too soon).

Full detail in the repository

- Mentor / student prompts: prompts/mentor/current.txt, prompts/student/current.txt
- Tools implementation: orchestration/tools.py, orchestration/relay.py
- Full best-run transcript: courses/prompt-engineering/logs/run_018/transcript.txt
- Full 5-section writeup incl. the complete dialog: docs/final_submission.md
- Change history: CHANGELOG.md